Hamid Abbasi

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Work Experience

Research Associate, London, Imperial College London *July 2025 – Present*

- Project **CASTLE** (Contrail Analysis using Stochasticity and Thermodynamic consistency in a Lagrangian Environment): Architected a scalable Lagrangian model of contrail-cirrus atmospheric interactions in C++ and Python, leveraging hybrid parallelization (MPI/OpenMP) on HPC cluster across distributed nodes; addressing the physically convoluted and computationally expensive problem of regional-scale contrail modelling. *Github repo currently private pending publication; source code available upon request.*
- Developed a novel stochastic model for cirrus cloud formation, explicitly linked to probabilistic parameterizations of atmospheric gravity waves to achieve an accurate simulation of the atmospheric ice life-cycle.
- Project **CLAMP**: Optimized legacy Python code by orders of magnitude, reducing execution time through advanced vectorization and Numba compilation; creating a novel and highly efficient plume-scale representation of contrail dynamics. *Github repo currently private pending publication; source code available upon request.*

Graduate Teaching Assistant, Manchester, University of Manchester *Sep 2022 – Jan 2024*

- Process Design: Teaching and helping students with construction of P&ID designs
- Engineering Math 2 & 3: Linear algebra, ODEs & PDEs, Multi variable Calculus, Complex Analysis
- Fundamentals of Thermodynamics: Analysis of 1st and 2nd laws in various thermal systems.

Lecturer, Manchester, Wanhua Chemical Group *Sep 2023 – Jan 2024*

- Reaction Systems: Conducted sessions on the application of Monte Carlo methods (Gillespie algorithm), in the context of simulating the stochastic time-dependent dynamics of chemical reactions.

Graduate Teaching Assistant, Iran University of Science and Technology *Sep 2018 – Jan 2021*

- Optimization of Thermal Systems; Thermodynamics and CFD; Advanced IC Engines; Fluid Mechanics 1 & 2

Education

University of Manchester, *PhD in Chemical Engineering* *Jan 2022 – Jul 2025*

- **Thesis:** multiscale model of solid oxide fuel cells /SOFC-Project
- Awarded the University of Manchester's flagship scholarship (President's Doctoral Scholarship)
- Designed a parallelized finite-volume model in Python to model and solve nonlinear PDEs at different length scales.
- Developed an integrated pore-scale, and cell-scale model to simulate chemical reactions involved in the internal reforming of hydrocarbon fuels within porous electrodes of SOFC accounting for chemical non-equilibrium.

Iran University of Science and Technology, *MSc in Mechanical Engineering* *Sep 2017 – June 2020*

- GPA: 3.88/4.0 (Official transcript); Ranked 1st among 2020 graduates.
- **Thesis:** Developed a numerical model to solve PDEs governing transient turbulent airflow inside air compressors.

Shiraz University, *BSc in Mechanical Engineering* *Sep 2013 – June 2017*

- GPA: 3.55/4.0 (Official transcript) Among top 10% of graduating students.

Selected Publications

Full list: Google Scholar; h-index=10, Citations=829; Published 16 articles in leading journals, 11 first-author and 6 conferences; International collaborations with EPFL Switzerland, UBC Canada, UNSW Sydney, University of Leeds

- A Pore-Scale Computational Framework for Enhancing Solid Oxide Fuel Cell Performance through the Design of Anode Porous Configurations; 2025, ***H Abbasi****, et al.* 10.1149/1945-7111/adbfc4
- Multiscale Modeling of Solid Oxide Fuel Cells Using Various Microscale Domains Across the Length of the Cell; 2024, ***H Abbasi****, et al.* 10.1016/B978-0-443-28824-1.50387-2
- Application of machine learning tools for constrained multi-objective optimization of an HCCI engine; 2021 *A Ghareghani,* ***H Abbasi****, P Alizadeh.* 10.1016/j.energy.2021.121106
- Thermal optimization of shell and tube heat exchanger using porous baffles; 2020, *MH Mohammadi,* ***H Abbasi****, et al.* 10.1016/j.applthermaleng.2020.115005
- Shape optimization of segmental porous baffles for enhanced thermo-hydraulic performance of shell-and-tube heat exchanger; 2020, ***H Abbasi****, et al.* 10.1016/j.applthermaleng.2020.115835

Certificates

nVIDIA: Accelerating CUDA C++ Applications with Multiple GPUs (Link), Fundamentals of Accelerated Computing with CUDA Python (Link)

- Utilized all available GPUs on a single node to scale computation-intensive workloads.
- Gained experience in GPU-accelerated High Performance Computing (HPC) with CUDA Python using Numba, covering custom kernels, memory coalescing, shared memory, multi-dimensional grids.

nVIDIA: Deep Learning (Link), and Transformer-Based Natural Language Processing (Link)

- Built, feature-engineered, and fine-tuned deep learning models with PyTorch, covering CNN, efficient transfer learning techniques (from pre-trained models), data augmentation, and natural language processing.

nVIDIA: Accelerating End-to-End Data Science Workflows (Link)

- Used GPU-accelerated framework RAPIDS (cuDF, cuML, dask) to explore causal relationships in healthcare data, and create interactive visualizations from large datasets uncovering insights in complex systems.

HarvardX: Data Science: Probability, Inference and Modeling, Data Wrangling, Linear Regression, Machine Learning

- Applied statistical inference, linear regression, and probability theory to solve real-world challenges. Gained hands-on experience in confidence intervals, Bayesian modeling, and predictive analytics through practical projects.
- Developed interactive dashboards to explore multiple real-world problems using advanced statistics and probability. See /Poll-Simulation, /Loan-Risk-Analysis, and /Bayesian-Inference

Kaggle: Intro to Deep Learning, Intro and Intermediate Machine Learning, and Time Series

- Built and optimized DNNs and ensemble models (Random Forest, XGBoost) on real-world datasets, applying techniques like dropout, batch normalization, and cross-validation to prevent overfitting and ensure robust pipelines.

University of Manchester at Harwell: Visualisation and Quantification of Tomographic Datasets

- Hands-on experience with tomographic data analysis, combining theoretical insights and software experience.

Selected Projects

Contrail/Atmospheric Models *2025-2026*

- **Project CASTLE:** Developed a highly scalable, multi-scale Lagrangian framework for contrail modeling, designed for seamless integration with a diverse range of plume-scale contrail sub-models.
- **Project CLAMP:** Developed a high-performance Lagrangian model of plume-scale contrail dynamics.

Statistics and Probabilistic Models *2024*

- /Physics-Informed-ML: Created an interactive tool using PyTorch to model nonlinear dynamics of Van der Pol oscillator using a Tensor Basis Neural Network (TBNN), allowing real-time training based on user inputs.
- /HMM-synthetic: Designed an interactive dashboard to generate synthetic time series data with specific Markov properties, leveraging Hidden Markov Models to evaluate and analyze its performance on these simulated datasets.
- /Network-Autoregressive-Model: Developed a Python-based tool to simulate and estimate discrete-time linear autoregressive models for networked systems, incorporating adjacency constraints, external inputs, and noise.
- /ML-Toolbox-Finance: Developed efficient pipelines in Python to quickly evaluate the performance of different classes of ML models (Deep Learning, Hidden Markov Model, Random Forest) forecasting an asset's performance.
- /Markov-Chain : Built an interactive dashboard in Python to visualize Markov transition matrices of assets.

Quantitative Finance *2024*

- /Black-Scholes-Model: Created an interactive dashboard explaining the derivation of Black-Scholes model and extensions like implied volatility and American option pricing.
- /Stochastic-Processes: Developed a dashboard for Portfolio Management: Applied Modern Portfolio Theory to assess the efficient portfolio. Evaluated the Value at Risk (VaR) of each portfolio within a given confidence level.
- Algorithmic Trading: Designed and backtested different trading strategies based on statistical /Pair-Arbitrage, range Breakout and retest (/Adaptive-Channel-Strategy), and /Zero-Intelligence-Strategy

Technologies

Programming: Python, **C++**, **MATLAB**, **Fortran**, LaTeX, *Parallel Computing* (MPI, OpenMP, mpi4py, CUDA, RAPIDS, dask), *High-Dimensional Data Analysis* (Pandas, Polars, Xarray)

Physics Simulation Package: COMSOL, ANSYS